Neexdzii Kwah Restoration Planning

Executive Summary

S E R N B C

SOCIETY FOR ECOSYSTEM RESTORATION
IN NORTHERN BRITISH COLUMBIA

Version 0.2.4 DRAFT I 2026-03-16

Prepared for the Wet'suwet'en Treaty Office Society

Prepared by Al Irvine, R.P.Bio — New Graph Environment Ltd.

on behalf of the Society for Ecosystem Restoration in Northern British Columbia

Full Report I Source Code and Data I Changelog I Latest Executive Summary

Claude Opus 4.6 (Anthropic) assisted with analysis, writing, and code development. All scientific interpretation and conclusions are the responsibility of the authors.

Executive Summary

This report presents a governance and prioritization framework for restoration of the Neexdzii Kwah (Upper Bulkley River) watershed, grounded in Wet'suwet'en stewardship values and supported by an integrated suite of open-source spatial analysis tools. The framework is not a static plan — it is a living system designed to evolve as understanding deepens and new data becomes available.

Governance and prioritization. The prioritization framework proposes a four-tier governance structure — Stewardship Council, Technical Working Group, Implementation Partnerships, and Community Roundtable — that places Wet'suwet'en direction at the centre of restoration decision-making. Three diagnostic gates filter candidate sites before investment: certainty of diagnosis, whether active degradation is ongoing, and access and willingness to act. Scoring principles were developed from community workshops where participants ranked restoration parameters. Workshop groups converged on the same four priorities — Cultural Significance, Probability of Success, Watershed Function, and Proximity to High-Value Fish Habitat — though the relative weighting differed across groups. These perspectives are built into the weighting system. The framework acknowledges that habitat restoration alone cannot deliver salmon recovery. Skeena chinook and coho face sustained declines driven by fishing pressure — both marine and in-river — that exceeds what these populations can sustain. Reducing harvest is the necessary precondition; without it, restored habitat produces fish that are removed before they can spawn. Rebuilding habitat capacity now ensures the watershed can support recovery when harvest rates are brought in line with current productivity.

Sub-basin delineation. The watershed was divided into 14 sub-basins to enable prioritization and facilitate communication of issues and opportunities within each area. Natural break points — Bulkley Falls and Buck Falls — define the primary divisions due to their significance as barriers to salmon access. Within these regions, sub-basins were delineated based on groupings of habitat and landscape characteristics at comparable scales, using FWA-referenced spatial hydrology tools via `fresh` and the interactive `breaks` sub-basin delineation tool. Each sub-basin integrates fish habitat modelling, land ownership, reserve lands, cultural sites, and land cover change into a single characterization that supports comparison and investment targeting.

Floodplain modelling and land cover change. The functional floodplain — where the river does geomorphic work including sediment storage, nutrient exchange, riparian recruitment, and large woody debris delivery — was modelled using the Valley Confinement Algorithm implemented in `flooded`. Within the modelled floodplain extent, satellite-derived land cover classification (2017–2023) using `drift` reveals a consistent pattern of tree cover loss and agriculture expansion. The magnitude of detected change depends on the stream network used to define the floodplain model: using 4th order and larger streams modelled as coho accessible yields approximately 680 ha of net tree cover loss within the Neexdzi Kwah watershed, while expanding to 3rd order streams anchored to waterbodies connected to that network by 1st order tributaries increases the estimate to over 1,000 ha. Both are valid representations at different scales — the range reflects real sensitivity to

model assumptions and underscores why documenting those assumptions matters. Individual sub-basin tree loss rates range from -1% to -22%, with agriculture expansion up to 59% in the most affected areas. Ongoing work will further quantify these scenarios to visualize the different stream networks, resulting floodplain extents, and land cover outcomes.

Historic aerial photographs. Nearly 10,000 provincial aerial photographs (1963–2019) have been downloaded and georeferenced using `fly`. This collection represents the entire provincial dataset where photograph footprints land within the Neexdzi Kwah watershed. It is served as a queryable spatial catalog — available programmatically and from within GIS systems such as QGIS. The interactive `diggs` application allows users to select photos based on centroid and estimated footprint overlap with custom areas of interest (ex. modelled floodplain extents). This 50-year visual record allows direct comparison of historic and current conditions at any location in the watershed — building understanding of how the landscape has changed over time and grounding restoration targets in documented prior condition.

Site assessments. Field reviews at 26 locations over 2024–2025 documented conditions at past restoration sites, historic prescription locations, and newly proposed areas. A recurring theme was riparian and floodplain vegetation removal, channelization of streams and cattle-related degradation at scales far exceeding the footprint of past restoration investments. At sites where past work has been completed, riparian buffer widths were narrower than what current understanding suggests is needed for effective recovery. Road and rail erosion protection consistently lacked vegetated riprap, soft armouring, and effective riparian buffers. UAV flights at the majority of sites provide high-resolution baseline imagery for future monitoring, cataloged via `stac_uav_bc`. These observations feed directly into the prioritization framework and provide the baseline against which future restoration effectiveness will be measured.

Collaborative GIS. A shared QGIS project integrating provincial background layers, field data collection forms, spatialized historic restoration prescriptions, and all analysis outputs serves as the spatial integration layer for this work. Through cloud-based mobile GIS, the full project travels into the field on mobile devices — functioning as both the primary navigation system during site visits and the platform for standardized data collection. The ability to overlay fish habitat modelling, historic prescriptions, floodplain change detection, land tenure, and traditional knowledge in a single environment allows users to understand the spatial context of each location and make informed decisions about restoration priorities. Critically, this is not just an analytical tool — it is the primary platform for field data collection and the spatial integration layer that connects all other project components.

Aquatic health monitoring. Benthic invertebrate sampling was conducted at three sites on the Neexdzii Kwah mainstem within high-value chinook spawning and rearing habitat, with triplicate kick samples at each site to enable statistical distinction between real site differences and natural variability. Full taxonomic analysis, interpretation, and site comparisons are presented in a

dedicated report, with data to be uploaded to the federal CABIN database. The benthic report also compiles approximately 50 years of water quality data from 13 provincial Environmental Monitoring System stations within the watershed, evaluating nutrient conditions and point-source influences against provincial and federal guidelines and benchmarks.

Environmental DNA (eDNA) sampling targeting chinook, coho, sockeye, and bull trout was conducted below Bulkley Falls and approximately 1 km downstream to detect species presence within the reach at the time of collection, with analysis pending from the University of Northern British Columbia.

Tools as deliverables. The spatial analysis tools developed for this work are open-source R packages designed to work at any scale and in any watershed in the province:

- Stream network spatial hydrology — `fresh`
- Interactive sub-basin delineation — `breaks`
- Floodplain delineation — `flooded`
- Land cover change detection — `drift`
- Airphoto footprint estimation and optimal coverage selection — `fly`
- Interactive airphoto explorer — `diggs`
- Provincial LiDAR digital elevation models (DEMs) — `stac_dem_bc`
- UAV imagery catalog — `stac_uav_bc`
- Historic airphoto catalog — `stac_airphoto_bc`

These tools were developed across multiple projects and through ongoing research investment — they are not single-use products tied to any one initiative. The capacity they build — for ourselves and for others — to understand and manage conservation and restoration is as much a deliverable as the analysis results themselves.

This report is a living document, with ongoing revisions and updates tracked in the project changelog. The research and monitoring programmes described here are not just technical exercises — they are acts of territorial stewardship, grounded in the understanding that health, ecosystems, social systems, and watershed management must be governed together (Harris 2011). The United Nations Decade on Ecosystem Restoration (2021–2030) reinforces these principles through international standards of practice that call for addressing root causes of degradation, benefiting the people with the deepest connection to the ecosystem, and integrating governance across scales (FAO et al. 2023). We are humbled by the depth of knowledge that exists for this region, both through traditional knowledge and scientific study. Recovery efforts must go beyond technical interventions and include a commitment to learning from the vast body of knowledge held by the Wet'suwet'en people — including cultural practices, oral tradition (Kungax), Wet'suwet'en Law (Ink Nu'at'en), and the deep connections between families, house groups, and the lands and waters that sustain them (University of British Columbia Library, n.d.; Morin 2016).

References

FAO, SER, and IUCN CEM. 2023. *Standards of Practice to Guide Ecosystem Restoration: A Contribution to the United Nations Decade on Ecosystem Restoration. Summary Report*. FAO. https://doi.org/10.4060/cc5223en.

Harris, Sandra Martin. 2011. "Yinka Dinii Ha Ba Ten, the Ways of the People of the Earth: A Social-Ecological Framework for Sustainability." Master's thesis, University of Northern British Columbia. https://unbc.arcabc.ca/islandora/object/unbc%3A16575?solr_nav%5Bid%5D=211239513a91efe5652d&solr_nav%5Bpage%5D=0&solr_nav%5Boffset%5D=0.

Morin, Mélanie H. 2016. *Niwhts'ide'ni Hibi'it'ën: The Ways Of Our Ancestors. Witsuwit'en History & Culture Throughout The Millennia*. 2nd ed. School District #54 (Bulkley Valley) and Witsuwit'en Language and Culture Authority. https://www.sd54.bc.ca/contact-us/.

University of British Columbia Library. n.d. "Delgamuukw Trial Transcripts Collection." Accessed February 19, 2025. https://open.library.ubc.ca/collections/delgamuukw.